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Abstract

Sustainable Development Goal 1's (SDG1) commitment to ending poverty cannot be separated from the structural terms on which the global energy transition is being financed. This paper examines how accelerating demand for clean energy transition minerals (CETMs) like lithium, cobalt, nickel, graphite, and copper requires strategic development to alleviate poverty in mineral-exporting economies of the Global South. Drawing on economic complexity theory and trade-carbon-inequity frameworks, research argues persistent poverty in these economies is driven in part by systematic value loss at the transition between raw extraction and midstream processing, which is where most supply-chain value-complexity gains occur, yet largely uncaptured by minerally endowed developing economies. Existing institutions, notably the Extractive Industries Transparency Initiative (EITI), address revenue disclosure but not downstream allocation, leaving an institutional vacuum this paper identifies as a structural driver of poverty. Using case evidence from the Democratic Republic Congo's (DRC) cobalt sector and Zambia's copper sector, the paper proposes three complementary SDG1 target extensions, a Domestic Value-Added Retention Ratio (DVARR), a Reinvestment Compliance Rate (RCR), and a Processing Capacity Utilization Rate (PCUR), designed to make value capture and reinvestment measurable in ways current indicators cannot.

Introduction

The IEA (2022) reports that global demand for clean energy transition minerals is accelerating, and countries supplying them must invest in domestic ore processing to become the countries that stand to gain most from that demand. This is a matter of where value is captured as extraction volume in these economies is high and rising, but the wealth it generates is not (Braunstein & Chuchko, 2025). This paper argues that persistent poverty in critical-mineral-exporting economies stems from a specific point of loss in the transition between raw extraction and midstream processing, such that these economies are structurally prevented from capturing the value their resources generate. Existing institutions like the EITI governs disclosure of revenue once it exists but says nothing about whether that revenue is reinvested or whether processing capacity ever gets built. The result is an institutional vacuum at exactly the stage where value is created and, is necessary to consider how to equip SDG1 targets to measure. Using DRC and Zambia case evidence, this paper proposes three complementary extensions to SDG1 targets to make value

retention, reinvestment, and processing capacity measurable, and closes with a personal statement of intent.

Current scientific understanding and policy approaches and instruments

The global clean energy transition is deeply intertwined with mining. Processing and manufacturing of CETMs occurs within a handful of technologically advanced economies, while raw-material extraction remains disproportionately located in the Global South (Nsude et al., 2024). The IEA (2022) projects CETM demand to roughly double from 2026 levels by 2040 under its Sustainable Development Scenario, with lithium demand rising to approximately 42 times 2020 levels, graphite 25 times, cobalt 22 times, and nickel 19 times, while copper nearly triples (Nsude et al., 2024, p. 403). Fewer than twenty countries dominate production, notably the DRC in cobalt and China in rare earths, meaning a handful of economies bear a disproportionate share of both extraction burden and buyer-concentration exposure, while much of the associated environmental and social damage is absorbed by the Global South even as cleaner air and production accrue to the Global North (Sovacool et al., 2020, p. 17).

Recent literature formalizes why this demand surge fails to raise living standards in exporting countries. Ren et al. (2025) find trade-related carbon inequity escalates under the EU's Carbon Border Adjustment Mechanism absent instruments like revenue recycling. Li et al. (2026), applying economic complexity theory to the CETM supply chain, find that entrenched extraction-stage strength creates path dependency rather than a foundation for upgrading; 76 percent of total value-complexity gain across these supply chains occurs specifically at the transition from raw material to intermediate processing, meaning mineral-endowed nations require capital investment in domestic processing infrastructure to pursue equity. Raw-material complexity is also geologically bimodal, since some minerals offer a lower ceiling on extraction-stage value regardless of policy.

Existing frameworks address only dimensions of this imbalance. The EITI sets a global disclosure standard for extraction-sector revenue, exploration, licensing, and allocation (Pafadnam, 2024, p. 2), and membership is associated with increases in FDI and revenue collected, the two channels through which it appears to affect growth (Pafadnam, 2024), but its mandate stops at disclosure rather than downstream allocation, leaving an institutional vacuum that is itself a structural driver of the value-capture gap. In Nigeria, this vacuum is compounded by domestic political economy

where the EITI's disclosure model was absorbed into existing power asymmetries such that disclosure did not translate into accountability (Moses et al., 2024). Similarly, Hilson et al. (2016) found ethical-minerals scheme designers often lack sufficient local context to reach artisanal producers, calling for integrating informal operators into scheme design rather than formalizing around them, as with EITI, improving visibility without altering the terms on which value is captured.

Suggested new/complementary goals and indicators to move the challenge to a new level

Considering SDG1 targets within the United Nations Global Goals, the following three extensions build on the targets more specifically rather than replace them[1]. The first, extending Target 1.4, would commit signatories to increasing the export-revenue share mineral-exporting economies retain through domestic midstream processing to address the transition point Li et al. (2026) identify as capturing most value-complexity gain. A Domestic Value-Added Retention Ratio (DVARR) would measure the proportion of export value attributable to in-country processing relative to raw extraction value, tracking composition rather than volume. The IEA (2026) reports Latin America and the Caribbean, facing a gap like Zambia and the DRC, currently refining only about one-fifth of mined output, and estimates localizing processing of two-thirds of regional copper could raise value by nearly 50%, to roughly 220 billion USD by 2035 (p. 13). The DRC remains dominant in cobalt mining even as refining stays marginal, while South Africa has revised its strategy toward domestic beneficiation, showing mining dominance and value capture are not equivalent. If processing investment materializes, this upgrading would convert a meaningful share of that opportunity into wages, tax base, and reinvestment revenue, which the DVARR is positioned to register directly.

Second, extending Target 1.B, responds to Moses et al. (2024), who find NEITI's disclosure model in Nigeria was absorbed into existing power asymmetries without producing accountability. The Reinvestment Compliance Rate (RCR) extends disclosure to verify whether revenue reaches

[1] United Nations (n.d.-a) Target 1.4: By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land and other forms of property, inheritance, natural resources, appropriate new technology and financial services, including microfinance. Target 1.A: Ensure significant mobilization of resources from a variety of sources, including through enhanced development cooperation, in order to provide adequate and predictable means for developing countries, in particular least developed countries, to implement programmes and policies to end poverty in all its dimensions. Target 1.B: Create sound policy frameworks at the national, regional and international levels, based on pro-poor and gender-sensitive development strategies, to support accelerated investment in poverty eradication actions.

infrastructure or social protection rather than treasury, though it could itself be captured unless verification sits outside that hierarchy. Pafadnam (2024) confirms the EITI mandate covers disclosure of allocation and resulting spending, but not enforcement over use. His valid transmission channels, foreign direct investment and resource revenue mobilization, are upstream capture measures, with effects vanishing long-run and no significant corruption impact, consistent with EITI improving inflow, not what happens afterward. This burden falls unevenly, since EITI obligates governments and large companies, not individual operators. Hilson et al. document this in schemes like Fairtrade, where designers target "low-hanging fruit" since requirements are "beyond the capacity of many artisanal and small-scale mining (ASM) enterprises." Informal ASM, already excluded from the licensed sector, is excluded from this apparatus too as producers operate inside it, miners remain outside by the same exclusion. Whether this reduces poverty long-run depends on whether captured rents are reinvested domestically rather than absorbed into the pattern Pafadnam's results suggest, which the RCR aims to track.

Third, applying a Processing Capacity Utilization Rate (PCUR) to Target 1.A addresses the capability path dependency Li et al. (2026) identify as structural. Countries with entrenched strength in raw-material extraction face structural barriers to advancing into processing, precisely because that extractive strength creates path dependency rather than a foundation for upgrading, requiring capital investment in domestic processing infrastructure to work toward distributional equity. This dynamic manifests concretely as physical capacity constraints in the DRC. The DRC has moved from waiver-based restriction, with bans in 2013, 2019, and 2023 each accompanied by waivers where domestic smelting capacity was insufficient, to a full concentrate export ban effective June 29, 2026, though one-year waivers may still be granted in strategic circumstances (Kasongo & Adombila, 2026). A PCUR measuring smelting capacity relative to raw ore output would make these capability gaps visible in a way EITI-style, disclosure-focused indicators cannot, supporting concessional capital mobilization for midstream infrastructure. At Kamoa-Kakula, the DRC's largest copper mine and the fourth largest in the world by 2024 production, seismic activity caused severe flooding and infrastructure damage in 2025, cutting production by roughly one-third relative to prior guidance, with full recovery not expected until 2028 (IEA, 2026, p. 127). Outcomes are more favourable in the long-run if processing investment materializes, as in Indonesia's nickel sector, but the DRC's repeated policy reversals since 2013 undermine the

investor confidence long-horizon capital requires, reinforcing the need to pair export restrictions with sequenced, financing-linked implementation.

Statement of intent

I am deeply interested in energy economics, which overlaps with mining, trade, security, and ethics. I am drawn to cases like Zambia's copper sector and the DRC's cobalt sector, wondering why some resource-rich economies translate that endowment into durable prosperity while others remain structurally vulnerable. My sustainability lens, developed in ENV222, rejects one-size-fits-all solutions from Loring's (2020) "no panaceas" concept and favours context-specific analysis attentive to who captures value, who bears the cost, and whose knowledge shapes the decision. A standard cost-benefit read of Zambia's duty or the DRC's ban would register export revenue and my lens further asks who retains that value, and whether relief measures like smelter waivers establish dependencies producing harm decades later. With A.I. now driving demand for the same minerals CETM needs, who controls processing capacity and who captures value will only matter more. Although I can't design a smelter, I see myself working as an intermediary among disciplines like economics, engineering, and politics analysing data to build financing proposals that take accountability and intergenerational consequences seriously.

Conclusion

Environmentally just clean energy transition must factor how it affects those vulnerable to poverty. Rising extraction volume can coexist indefinitely with stagnant domestic value capture, because nothing in current mineral governance requires the two to move together. The DVARR, RCR, and PCUR are measurement tools aimed to achieve existing targets within SDG1, since export growth and disclosure compliance do not necessarily track whether value is retained or reinvested. Export restrictions can force domestic processing onto the agenda, but without financing sequenced to match engineering timelines, they risk destabilizing the communities that the SDG1 targets are meant to protect. Moreover, AI-driven computing infrastructure is now adding a second, rapid-growing source of CETM demand meaning. institutions that leave value capture unmeasured under one source of demand will leave it unmeasured under two. Who controls processing capacity and who captures value will only grow more consequential, and enacting policy to build a foundation that works toward alleviating these issues is a task for the interdisciplinary intermediaries.

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